sas innovate

Mapping Spatial Data with SAS Code

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#SASinnovate

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Abstract

Location is at the center of everything that we do. In today's data-driven landscape, understanding spatial data is essential for uncovering hidden insights and making informed decisions. Whether you deal with addresses, sales districts, administrative boundaries, or something else entirely, chances are high that there is a spatial element in your data. Spatial data provides crucial geographical context, enabling you to detect patterns, trends, and relationships that traditional data analysis methods might overlook.

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Accessing Spatial Data

Spatial data can come in a wide variety of formats. In this section, you import shapefiles as SAS tables using the MAPIMPORT procedure.

1. Open the **ImportingSpatialData.sas** program in SAS Studio.
   In SAS Studio, navigate to **SAS Content/SAS Innovate/Mapping Spatial Data** and double click **ImportingSpatialData.sas** to open it.

2. Examine the shapefile in File Explorer.
   a. On the Windows desktop screen, open the folder called **Lower48**.
      This is a shapefile of the continental United States. A shapefile is a collection of points. Points can either be independent of one another, connected as polylines, or connected as polygons.
   b. Examine the component files that make up the shapefile.
      The .shp file contains the geometry information that creates the polygons. The .shx file contains an index. The .dbf file contains attributes information, such as the state name. The .prj file contains projection information, and the .cpg file contains encoding information.

3. Upload the shapefile to a location accessible to SAS Viya.
   a. In order to import the shapefile, we need to make sure it's in a location accessible to SAS Viya.
   b. Open SAS Studio.
   c. On the SAS Server tab, select **Files/Home**.
   d. Click the **Upload** button.
   e. In the Upload Files window, click the **Add** button.
   f. Navigate to the **Lower48** folder on the **Desktop**.
   g. Select all of the shapefile components and click **Open**.
   h. Click **Upload**.
   i. Verify that all the shapefile component files were successfully uploaded.

4. Run the PROC MAPIMPORT step to import the **Lower48** shapefile.
   a. Examine the code.

```
proc mapimport datafile="/home/student/Lower48.shp"
               out=work.states;
run;
```

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The DATAFILE= option must include the full path to the location where we uploaded the shapefile. It must also contain all of the component files that make up the shapefile: the .shp file, the .shx file, the .dbf file, and the .prj file.

b. Highlight the PROC MAPIMPORT step and click **Run**.

5. Examine the imported shapefile.

   Click the **Output Data** tab.

   The imported **work.states** table has 166,263 rows – one for each coordinate point in the shapefile. Notice the **X** and **Y** columns, which contain the latitude and longitude coordinates of each point. The **SEGMENT** column identifies multipart polygons, including any state that has islands or is split into multiple pieces, such as Michigan or Hawaii. The rest of the columns come from the attribute table in the imported shapefile. For example, the **NAME** column uniquely identifies each polygon, as does the **GEOID** column. **GEOID** is a commonly used code that uniquely identifies all administrative areas for which the US Census Bureau reports data.

   Notice the **X** and **Y** values are -91 and 31.

6. Run the SGMAP procedure to plot the imported shapefile.

```
proc sgmap mapdata=work.states;
     choromap / mapid=name legendlabel="States";
run;
```

   This will create a simple outline map of the US states shapefile. In a later section, we will discuss the SGMAP procedure in more detail.

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Projecting Spatial Data

You might have noticed that that last map didn't look quite the way you might expect – like it was stretched all the northern border. That's because it is displaying unprojected data. The map of the US that we normally expect to see uses projected coordinates that remove some of that distortion by curving the northern border. Spatial data is a representation of the earth, which is a sphere. However, a map is flat, two-dimensional surface. Projecting spatial data refers to the process of converting spherical latitude and longitude coordinates to Cartesian coordinates that can be mapped on a flat surface.

1. Open the **ProjectingData.sas** program in SAS Studio.
   In SAS Studio, navigate to **SAS Content/SAS Innovate/Mapping Spatial Data with SAS Code** and double click **ProjectingData.sas** to open it.

2. Run the SGMAP procedure to see what the unprojected map data looks like again.
   ```
   title "US Map: Unprojected";
   proc sgmap mapdata=work.states noautolegend;
        choromap / mapid=name lineattrs=(color=blue);
   run;
   ```

   Notice that we use the NOAUTOLEGEND option to suppress the legend and the LINEATTRS= option to control the line color.

3. Examine the resulting map.
   Notice that the top border of the US is a straight line. This is because unprojected data appears distorted on a two-dimensional surface.

4. Run the PROC GPROJECT step to convert the map data set to the Albers' projection.
   ```
   proc gproject data=states out=states_proj
                 eastlong degrees;
      id name;
   run;
   ```
   We need to specify EASTLONG and DEGREES because the input data uses degrees and the longitude values increase in value moving east from the prime meridian. We use the ID statement to specify the ID value of each individual polygon.

5. Examine the new table on the Results tab.
   Notice that the **X** and **Y** values are now very different: 0.06 degrees instead of -91 degrees, and
   -0.098 degrees instead of 31 degrees. That is because we have now projected the

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data, which means that we have mathematically transformed the coordinate values.

6. Run the SGMAP procedure to see what the projected map data looks like.
```
title "US Map: Albers Projection";
proc sgmap mapdata=states_proj noautolegend;
     choromap / mapid=name lineattrs=(color=blue);
run;
```

7. Examine the map.
   Notice that in the projected map, the top border is now curved, and the map looks more like the country maps that we are used to. This projection does a much better job of maintaining the true shape of US than the unprojected data.

8. Save the program.
   Click the **Save** button.

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Creating a Map

Now that we have discussed how to access and manipulate spatial data, we're ready to take a closer look at how we can create attractive visualizations using SAS.

1. Open the **MappingSASInnovate.sas** program in SAS Studio.
   In SAS Studio, navigate to **SAS Content/SAS Innovate/Mapping Spatial Data with SAS Code** and double click **MappingSASInnovate.sas** to open it.

2. Examine the **MAPSGFK.US_COUNTIES** table.
   a. In the left pane, click the **Libraries** tab.
   b. Click the **MAPSGFK** library and then double-click **US_COUNTIES** to open it.
      SAS includes a number of map data sets that can be used to create maps. This is a map data set containing county polygons for the entire US. The **COUNTY** column uniquely identifies each county.

3. Run the DATA step to subset the data to just counties in Texas. The %INCLUDE statement creates two tables in the work library that we will use to add additional detail to this map.
```
%include "/home/student/workshop/VJB/TexasPop.sas";
data texas;
    set mapsgfk.us_counties;
    where statecode="TX";
run;
```

4. Create a simple outline map of Texas counties. PROC SGMAP builds maps in layers.
   a. The first step is to define the map data.
   b. The next step is to define the map background.
```
proc sgmap mapdata=texas;
     choromap / mapid=id;
run;
```
      The CHOROMAP statement specifies that we are creating a choropleth map. A choropleth map can take two data sources: the map (or spatial) data and the map response data being plotted. Since no response variable is being plotted, no additional data or options are needed.
   c. Highlight the code and click **Run**.
      The result is an outline of the counties in Texas. To take this to the next level, let's create a map shaded by population (also known as a *choropleth* map).

5. Examine the **TX_POP** table.

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a. In the left pane, click the **Libraries** tab.
b. Click the **WORK** library and then double-click **TX_POP** to open it.
This is a map data set containing the population of each county in Texas. Notice that it also a **FIPS3** column with a 3-digit code that matches the **COUNTY** column in the map data set. PROC SGMAP can use this as *map response data* to create a choropleth (shaded) map.

6. Create a population map of Texas.
   a. The first step is to define the map data. Specify the **WORK.TEXAS** table as the map data and the **WORK.TX_POP** table as the map response data.
   b. Next, add a CHOROMAP statement, specifying **POPULATION** as the response variable, and **FIPS3** as the ID variable in the map response data set.
   ```
   proc sgmap mapdata=work.texas
        maprespdata=work.tx_pop;
        choromap Population / id=fips3 mapid=county;
   run;
   ```
   c. Highlight the step and click **Run**.
   The resulting map displays the counties in Texas, shaded by population. Notice the map legend, which is continuous. Next, let's add the SAS Innovate 2026 location to this map.

7. Examine the **WORK.INNOVATE** table.
   a. In the left pane, click the **Libraries** tab.
   b. Click the **WORK** library and then double-click **INNOVATE** to open it.
   This is a map data set containing the location of Grapevine, Texas. Notice that it has **X** and **Y** columns with spatial coordinates.

8. Add a plot layer to the PROC SGMAP map highlighting this year's SAS Innovate location.
   a. The first step is to define the plot data, which in this case is **WORK.INNOVATE**. Notice the PROC SGMAP step is now using three different tables: the map (or spatial) data, the map response data, and the plot data.
   b. The next step is to add the background map. We can use the population map that we already created as a background.
   c. Next, use the SCATTER statement to add the **WORK.INNOVATE** locations. The X and Y argument specify the columns containing the coordinates.

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```
proc sgmap plotdata=work.innovate
           mapdata=work.texas
           maprespdata=work.tx_pop;
     choromap population / id=fips3 mapid=county;
     scatter x=x y=y;
run;
```

A coordinate appears on top of the choropleth map. However, it's hard to see and there's no context in the legend. Let's add a few finishing touches to make this map easier to use.

9. Enhance the scatter map by changing the scatter plot color, adding it to the legend, and adding a title.
   a. Add a GROUP= option and a MARKERATTRS= option.

```
title "Where is SAS Innovate?";
proc sgmap plotdata=work.innovate
           mapdata=work.texas
           maprespdata=work.tx_pop;
     choromap population / id=fips3 mapid=county;
     scatter x=x y=y / group=Conference
markerattrs=(color=black symbol=starfilled);
run;
title;
```

   The MARKERATTRS option allows you to add style attributes to control the appearance of the coordinate marker. THE GROUP option adds the value of the **Conference** column to the legend.

   b. Run the program.

   The resulting map contains a scatter map with easier-to-read markers, a choropleth population map, and OpenStreetMap base map.

10. Save the program.

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